

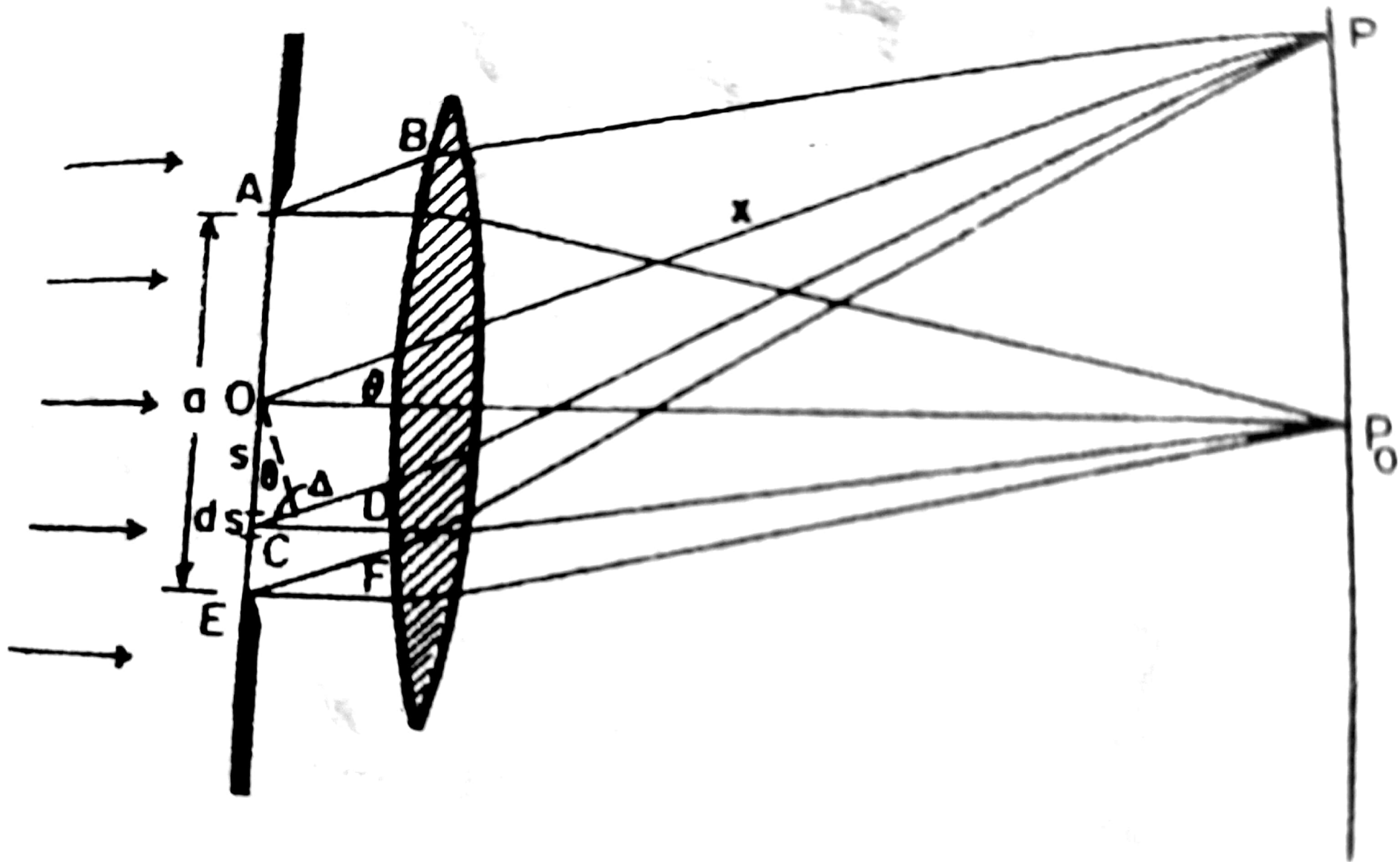


Fig. 28 2

on an opaque plate in which there is a slit. *A slit is a rectangular aperture of length large compared to its breadth.* The beam, transmitted through the slit, spreads out perpendicularly to the length of the slit. When this beam is brought to focus on a screen by a lens, a diffraction pattern is obtained. The pattern consists of a central band, much wider than the slit width, situated directly opposite to the slit and bordered by dark and bright bands of decreasing intensity. The central bright band is extremely intense and its width is twice as great as that of fainter side bands.

The origin of the pattern can be understood on the basis of interference of secondary wavelets. According to Huygens' principle, these wavelets can be thought of as sent out by every point of the wavefront at the instant it occupies the plane of the slit. A section of the slit, illuminated with parallel light from left, is represented in Fig. 28.2. Let a be the width of the slit and ds be an element of width of the wavefront in the plane of the slit, at a distance s from the centre O , which shall be referred to as the origin. Each secondary wavelet can be regarded as a spherical wave spreading out to the right. The part of each wavelet travelling normally to the slit, are brought to focus at P_0 by the lens while the parts, travelling at an angle θ with the normal are brought to focus at P . AB , CD , EF , etc. can, therefore, be regarded as parallel rays diffracted at an angle θ . Exactly along the same argument it follows that diffracted rays start from every point in every direction.

Let us first consider the wavelet emitted by the element ds situated at the origin. In spherical waves, the intensity decreases as the inverse square of the distance from the source. This follows directly from the fact that, provided there is no conversion of the energy into other forms, the same amount must pass through any sphere with the source as its centre. Since the area of a sphere increases as the square of its radius, the energy per unit area at a distance r from the source, or the intensity will vary as $\frac{1}{r^2}$. The

amplitude must then vary as $\frac{1}{r}$ and one may write the equation of a spherical wave as

$$y = \frac{a}{r} \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

where a is the amplitude at unit distance from the source. Thus the wavelet emitted by the element ds will produce an infinitesimal displacement dy at P_0 , which may be represented by the equation of a spherical wave in the form

$$dy = \frac{r.ds}{x} \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \quad (28.1)$$

where $r.ds$ is the amplitude of the wave at unit distance from the origin.

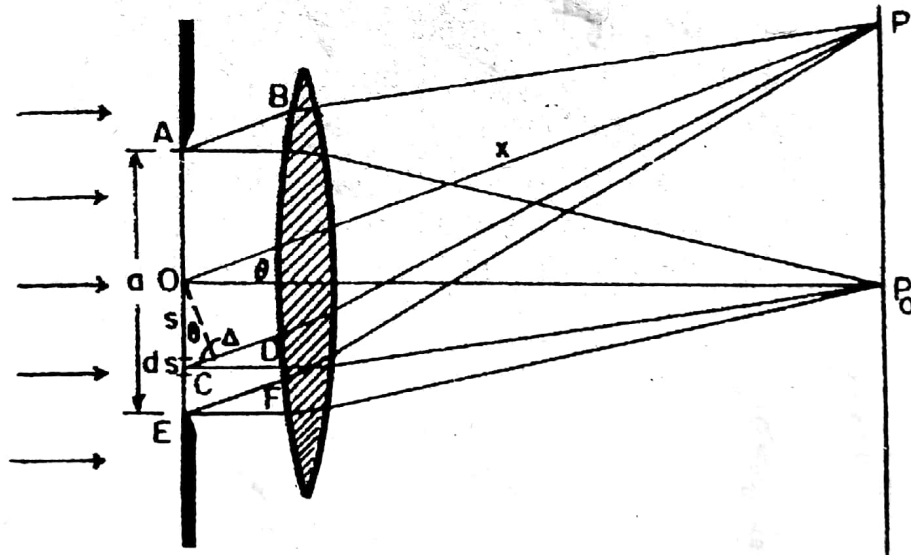


Fig. 28.2

The part of this wavelet reaching P traverses an optical path x_n and produces a vibration at P given by

$$dy_o = \frac{rds}{x} \sin 2\pi \left(\frac{t}{T} - \frac{x_n}{\lambda} \right) \quad (28.2)$$

As the position of ds is varied, the displacement it produces will vary in phase because of the different path length to P . When ds is at a distance s below the origin, the contribution can be seen from the figure to be given by

$$dy_s = \frac{rds}{x} \sin 2\pi \left(\frac{t}{T} - \frac{x_n + \Delta x}{\lambda} \right)$$

for Engineers

$$= \frac{rds}{x} \sin 2\pi \left(\frac{t}{T} - \frac{x_n}{\lambda} - \frac{s \sin \theta}{\lambda} \right) \quad (28.3)$$

where $\Delta x = s \sin \theta$ is the additional distance covered.

We shall now sum up the contributions of all elements ds from one edge of the slit to the other to get the resultant displacement at P . This sum is obtained by integrating the expression over the width of the slit, i.e., between the limits $s = -\frac{a}{2}$ and $s = +\frac{a}{2}$. During this summing up process the amplitude factor r/x may be dropped, since it is practically the same for all wavelets and one is interested only in *relative* intensities on the screen.

$$y = \int_{-\frac{a}{2}}^{+\frac{a}{2}} dy_s$$

$$= \int_{-a/2}^{+a/2} \sin 2\pi \left(\frac{t}{T} - \frac{x_n}{\lambda} - \frac{s \sin \theta}{\lambda} \right) ds \quad (28.4)$$

Assuming $\phi = 2\pi \left(\frac{t}{T} - \frac{x_n}{\lambda} \right)$ and $\psi = 2\pi s \frac{\sin \theta}{\lambda}$, we have

$$y = \int_{-a/2}^{+a/2} \sin (\phi - \psi) ds$$

$$= \int_{-a/2}^{+a/2} (\sin \phi \cos \psi - \cos \phi \sin \psi) ds$$

$$= \int_{-a/2}^{+a/2} \left[\sin \phi \cos \frac{2\pi \sin \theta}{\lambda} \cdot s - \cos \phi \sin \frac{2\pi \sin \theta}{\lambda} s \right] ds$$

now $\int \sin mx \, dx = -\cos \frac{mx}{m}$ and $\int \cos mx \, dx = \frac{\sin mx}{m}$

Hence

$$y = \left[\sin \phi \frac{\frac{2\pi \sin \theta}{\lambda} s}{\frac{2\pi \sin \theta}{\lambda}} + \cos \phi \frac{\frac{\cos 2\pi \sin \theta}{\lambda} s}{\frac{2\pi \sin \theta}{\lambda}} \right]_{-a/2}^{+a/2}$$

$$= \left[\frac{s \sin \psi \sin \phi}{\psi} + \frac{s \cos \psi \cos \phi}{\psi} \right]_{-a/2}^{+a/2}$$

when $s = +a/2$ $\psi = \frac{\pi a \sin \theta}{\lambda}$

$s = -a/2$ $\psi = -\frac{\pi a \sin \theta}{\lambda}$

$$\therefore y = \left[\frac{a}{2} \frac{\sin (\pi a \sin \theta)/\lambda}{(\pi a \sin \theta)/\lambda} \sin \phi + \frac{a}{2} \frac{\sin (\pi a \sin \theta)/\lambda}{(\pi a \sin \theta)/\lambda} \sin \phi \right.$$

$$\left. + \frac{a}{2} \frac{\cos (\pi a \sin \theta)/\lambda}{(\pi a \sin \theta)/\lambda} \cos \phi - \frac{a}{2} \frac{\cos (\pi a \sin \theta)/\lambda}{(\pi a \sin \theta)/\lambda} \cos \phi \right]$$

$$= a \frac{\sin (\pi a \sin \theta)/\lambda}{(\pi a \sin \theta)/\lambda} \sin \phi$$

$$= a \frac{\sin (\pi a \sin \theta)/\lambda}{(\pi a \sin \theta)/\lambda} \sin 2\pi \left(\frac{t}{T} - \frac{x_n}{\lambda} \right) \quad (28.5)$$

The above equation represents the resultant vibration at the point P . This represents a new simple periodic motion which has the same frequency as the source with a phase angle the same as the central ray (ray coming from the element ds located at the centre) and of amplitude

$$R = (a \sin \beta)/\beta \quad \text{where } \beta = (\pi/\lambda) a \sin \theta$$

Due to the neglect of the factor r/x during the summing up process, this expression for R differs by some arbitrary constant from the true amplitude. It will, therefore, be more correct to write

$$R = R_0 \frac{\sin \beta}{\beta} \quad (28.6)$$

The significance of the constant R_0 will appear below. The quantity β is a convenient variable, and can be expressed as

$$\beta = \frac{1}{2} \left(\frac{2\pi}{\lambda} \cdot a \sin \theta \right) \quad (28.7)$$

Now $a \sin \theta$ is the path difference of two rays originating from opposite edges of the slit. Hence β signifies one-half the phase difference in radians between the contributions from opposite edges of the slit. It determines the intensity I by the relation.

$$I = R^2 = R_0^2 \frac{\sin^2 \beta}{\beta^2}$$

$$I = R_0^2 \frac{\sin^2 \beta}{\beta^2} = k r_0^2 \quad (28.8)$$

Position of principal maxima

On writing the expansion of $\sin \beta$ in ascending powers of β , the expression for the resultant amplitude R becomes

$$R = \frac{R_0}{\beta} \left(\beta - \frac{\beta^3}{3!} + \frac{\beta^5}{5!} - \frac{\beta^7}{7!} + \dots \right) \quad \text{sin } \beta \text{ expansion}$$

$$= R_0 \left(1 - \frac{\beta^2}{3!} + \frac{\beta^4}{5!} - \frac{\beta^6}{7!} + \dots \right)$$

It is obvious that R will be maximum, if the negative terms vanish. This is only possible when

$$\beta = \frac{\pi a \sin \theta}{\lambda} = 0; \quad \text{or, } \theta = 0.$$

For $\beta = 0$, the quotient $\sin \beta / \beta$ becomes indeterminate but it should be remembered that $\sin \theta$ approaches θ for small angles and is equal to it when θ vanishes. Hence for $\beta = 0$ (i.e., $\theta = 0$), $R = R_0$; Thus the maximum value of R is R_0 ; it represents the amplitude when all the wavelets arrive in phase. The condition $\theta = 0$ simply means that this maximum is formed by parts of secondary wavelets which travel normally to the slit. R_0^2 is then the value of the maximum intensity which is designated by I_0 . The position of this maximum is, therefore, directly opposite to the slit and is at the

centre of the pattern. This maximum is known as the principal maximum and as explained below, it is bordered symmetrically by the dark and the bright bands.

Position of minimum intensity

The intensity will be zero in the diffraction pattern at the positions where $\sin \beta = 0$. The values of β which satisfies the above condition are $\beta = m \pi$ where $m = \pm 1, \pm 2, \pm 3, \dots$ which gives on substituting the value of β

$$\frac{a \pi \sin \theta}{\lambda} = \pm m \pi$$

or, $a \sin \theta = \pm m \lambda$.

where $m = 1, 2, 3,$

The value $m = 0$ is not admissible, because the value $\theta = 0$ corresponds to the principal maximum.

The first minimum occurs at

$$\theta = \pm \sin^{-1} \left(\frac{\lambda}{a} \right), \text{ the second minimum at}$$

$$\theta = \pm \sin^{-1} \left(\frac{2\lambda}{a} \right), \text{ etc. In words, the rays diffracted at an angle } \theta$$

interfere destructively if the path difference between the extreme diffracted rays is an integral-multiple of λ . Since $\sin \theta$ cannot exceed unity, the maximum value of m is the integer which is less than (and closest to) $\frac{a}{\lambda}$.

Position of secondary maxima

In addition to the principal maxima at $\theta = 0$ there are weak secondary maxima between the equally spaced minima. The exact positions of these maxima can be obtained by differentiating the expression for intensity $I = (I_0 \sin^2 \beta) / \beta^2$ with respect to β and equating it to zero. Therefore, for maxima and minima, we have the condition,

$$\frac{dI}{d\beta} = 2I_0 \left[\frac{\sin \beta \cos \beta}{\beta^2} - \frac{\sin^2 \beta}{\beta^3} \right] = 0$$

$$\text{or, } \sin \beta [\beta - \tan \beta] = 0$$

Hence, either

$$\sin \beta = 0 \quad \text{or, } \beta - \tan \beta = 0.$$

Now $\sin \beta = 0$ gives the values of β (except zero) for which the intensity is zero (positions of minimum intensity). The conditions of maxima are, therefore, given by the roots of the equation

$$\beta - \tan \beta = 0; \quad \text{or, } \beta = \tan \beta.$$

The above equation is a *transcendental* equation. The values of β satisfying this equation are given by the points of intersection of the curves

$$y = \beta \quad \text{and} \quad y = \tan \beta, \text{ plotted on the same graph.}$$

The curve $y = \beta$ is simply a straight line through the origin and inclined at 45° to the axis of β . To plot the curve $y = \tan \beta$, the following characteristics of the curve were derived mathematically.

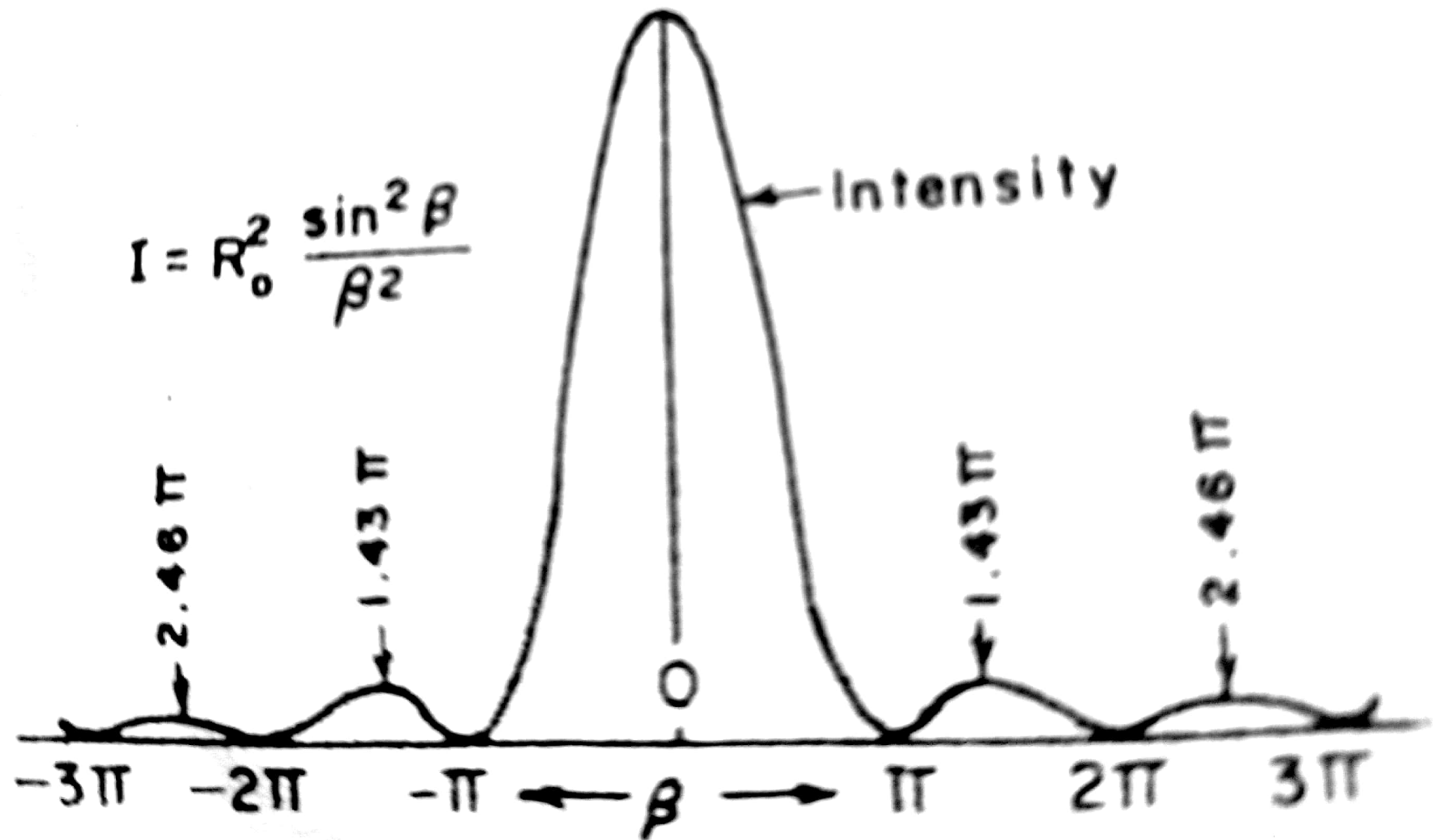
(i) The curve cuts the axis of β at points where $y = \tan \beta = 0$, that is, at points given by $\beta = 0, \pm \pi, \pm 2\pi, \pm 3\pi$, etc.

(ii) At points where it cuts the axis of β , the curve is inclined at 45° to the axis of β at those points.

(iii) Asymptotes of the curve are parallel straight lines defined by $\beta = \pm \pi/2, \pm 3\pi/2, \pm 5\pi/2$, etc.

The equation $y = \tan \beta$, therefore, represents a family of curves having the characteristics mentioned above. The curves are plotted in Fig. 28.3 along with the curve $y = \beta$. As can be seen from the figure, the intersections occur at $\beta = 0, \pm 1.43\pi (\simeq \frac{3\pi}{2}), \pm 2.46\pi$

$(\simeq \frac{5\pi}{2}), \pm 3.47\pi (\simeq \frac{7\pi}{2}), \pm 4.48\pi (\simeq \frac{9\pi}{2})$, etc. The value $\beta = 0$, of course, gives the position of principal maximum. The other values namely $\beta = 1.43\pi, 2.46\pi, 3.47\pi, 4.48\pi$, etc. give the respective



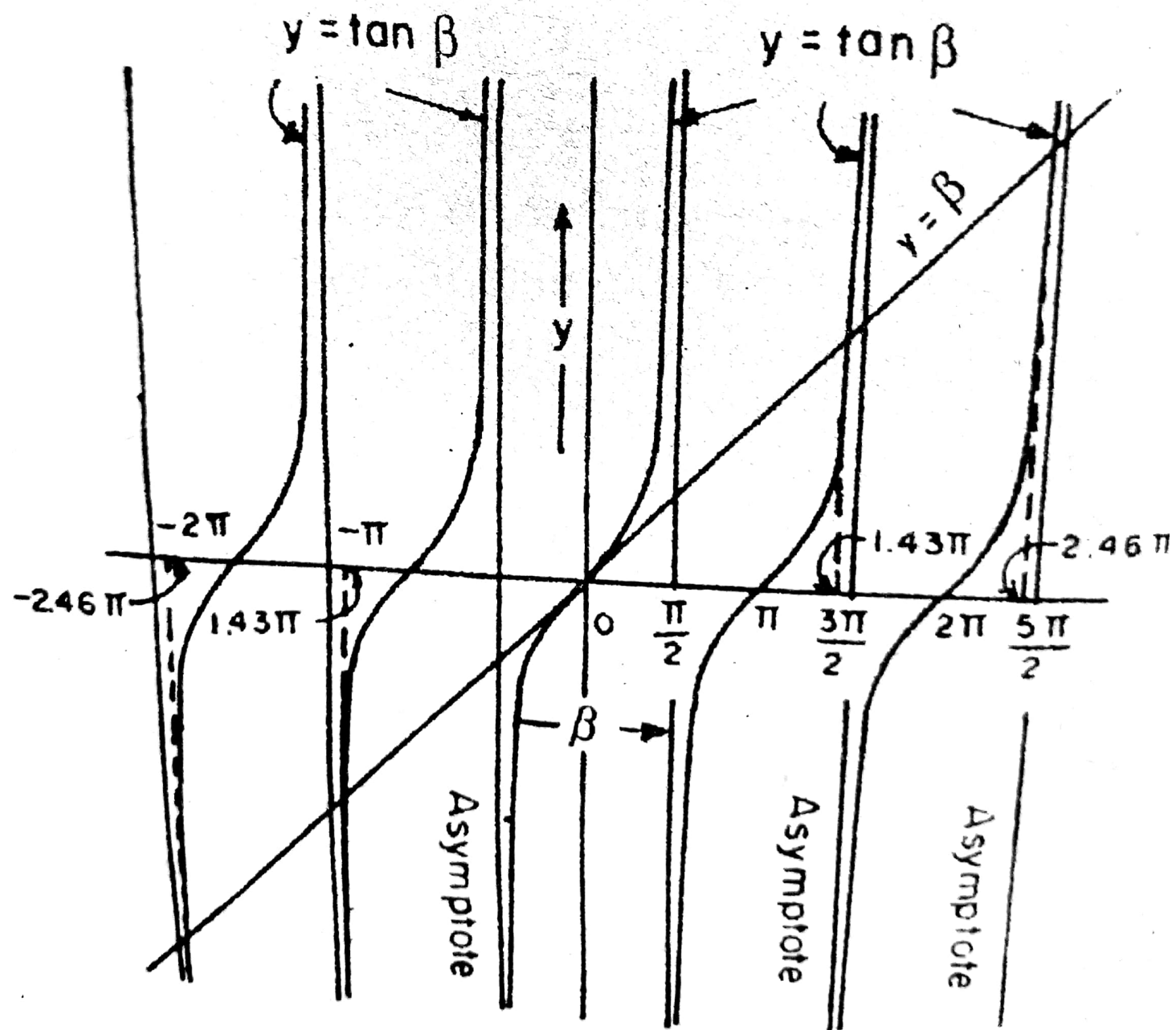


Fig. 28.3